Making Eating Out Less Energy Intensive

by Jonathan Livingston

Is there a way to increase energy efficiency and enhance productivity? That's the one question the Emerging Technologies group at Pacific Gas and Electric Company (PG&E) consistently asks regarding all facets of home and business life.

Nowhere is this question more applicable than in the food service industry—an industry that we all benefit from at least once in a while. Food service consumes more energy on a per square foot basis than any other commercial operation—$1 billion worth of energy a year in California alone. Restaurant equipment, such as ice makers and dishwashers, guzzle water and energy at an alarming rate.

FOOD SERVICE

In 1987, PG&E created the Food Service Technology Center (FSTC) to test food service appliances and determine baseline levels of water and energy use. Today the FSTC represents the gold standard in testing for commercial food service equipment; its 35 standard test methods are used to award Energy Star status to energy-efficient appliances. The FSTC provides technical training to other utilities and their food service customers such as facility managers, foodservice directors, and restaurant owners. Details of this program and other FSTC services are available at www.pge.com/fstc.

Corporations that have benefited from the FSTC's pioneering work include McDonald's, Safeway, and Taco Bell. Today's commercial kitchen managers save money, energy, and water by adopting high-efficiency solutions identified by the FSTC. That's good news for business and for the environment. To date, the FSTC is directly responsible for an estimated annual reduction of 1.3 billion kilowatt-hours of electricity and 30 million therms of gas. It still has a long way to go—about 80% of the $10 billion annual bill for commercial food service is wasted by the use of inefficient equipment.

Here are some of the high-efficiency appliances that have already been tested:

Low-rinse spray valves. Spray valves are used to spray food off dishware before it goes into the dishwasher. Today's high-efficiency spray valves are up to two-thirds more efficient than previous models, using as little as 1.6 gallons of water per minute (gpm). At only $60 per unit, this small investment can mean big savings. And the savings don't stop at the water meter: swapping one 5 gpm spray nozzle for a 1.6 gpm model saves over 1,800 therms annually, the equivalent of heating four homes for an entire year.

Connectionless steamers. A connectionless Energy Star steamer saves up to $3,000 per year in energy and $2,000 per year in water. Connectionless models do not require a drain, since they generate their own steam instead of sourcing from a stand-alone boiler (which is why they are called connectionless), and they do not require a separate supply of cooling water to cool the now-missing drain flow. These steamers include a small vent that allows any excess steam to seep into the open kitchen or to be sucked up into the overhead exhaust hood.

Air-cooled ice machines. Tests run by the FSTC have determined that a midsized (500-lb) water-cooled ice maker consumes up to 1,000 gallons of water every day that are never actually made into ice. This water is used to cool the unit, at an average annual cost of over $2,500. These tests have shown that air-cooled ice machines make more ice than water-cooled ice machines while cutting operating costs by up to 60%.

As an added bonus, air-cooled machines are quieter and cooler. PG&E's Emerging Technologies program continues to evaluate the energy savings that can be realized by high-efficiency ice makers. One avenue that the group is studying is whether ice can be made off-peak and stored for use during high-demand periods, in order to reduce demand during peak periods.

LAUNDRIES

Currently PG&E is looking for energy-saving solutions in large laundry facilities, such as those in hotels and hospitals. PG&E is investigating new ways laundry facilities can save energy and water by using ozone, which is a powerful oxidant and biocide. Ozone is created when air is exposed to ultraviolet light or electrical arcs. In an ozone laundry system, a generator injects ozone gas into laundry water. The highly oxidative properties of the ozone break the bonds between the soil and the fabric, eliminating the need for detergents and allowing laundry to be cleaned with shorter wash and dry cycles. A key potential benefit of ozonating systems is that ozone has a longer life in cold water than in hot water. Washing in cold water instead of hot water can save significant energy and money. Further savings potentially accrue from reduced chemical use, reduced labor costs, and longer linen life.

Jonathan Livingston manages the Emerging Technologies (ET) program in the Pacific Gas and Electric Company (PG&E) Customer Energy Efficiency Department.

For more information:

Visit www.pge.com/fstc to find out about rebates and other special incentives that are available to commercial kitchen managers and others who adopt the innovative technologies discussed in this article.